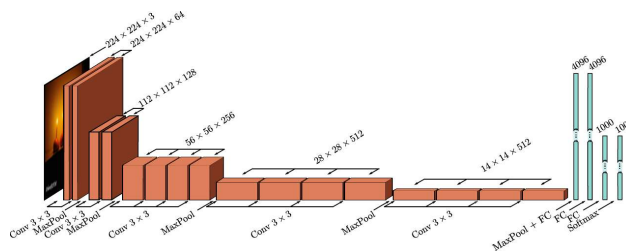
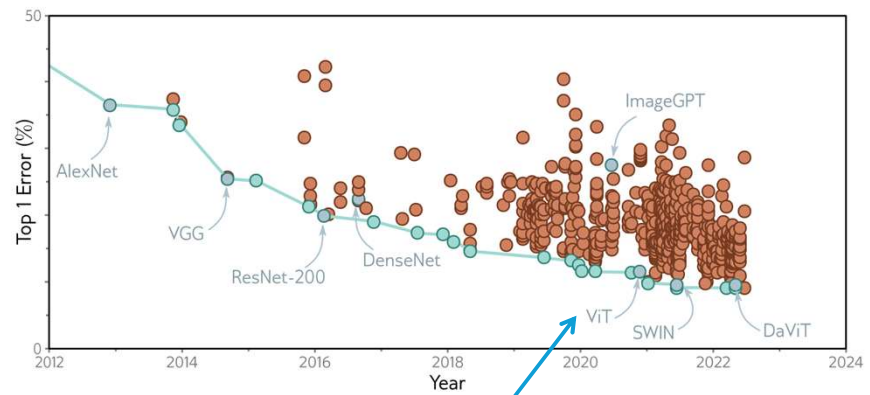


Overview

- CNN Architectures
- From CNN to Transformers

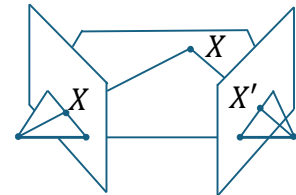


Vision Transformer exceeds human performance

1

Overview

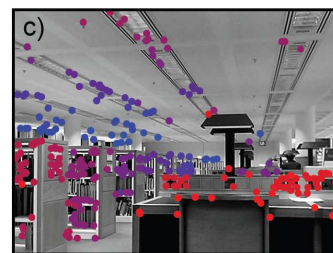
- How to do reconstruction if parameters are known
 - Unaligned but known cameras
- Some fun stuff!!
- Basics, buzzwords, and terminology on stereo
- The fundamental matrix F
 - Computing 'parameters' from F
- The essential matrix E
 - Computing 'parameters' from E



2

Story So Far

- The problem: compute depth
 - Cannot get (sparse) depth from single camera
- We can make progress with single camera
 - Obtain intrinsics and extrinsics parameters using a calibration pattern
 - 3D calibration pattern
 - 2D Calibration pattern (in practice)



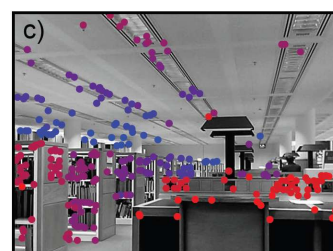
Sparse Depth

3

Story So Far



- We will use two ideas to get depth
 - A pair of views with two coupled cameras
 - Multiple views single known camera
- Eventually, multiple views with unknown cameras



Sparse Depth

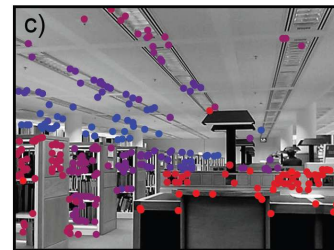
4

Story So Far

- The problem: compute sparse depth
 - Cannot get depth from single camera
 - Abandon: use 2 cameras
- Use a pair of views
 - Very specific camera with aligned optical image planes
- We perform calibration
- Depth computation is trivial *if pair-wise correspondence is known*



Stereo parameters: K and (R, T) known

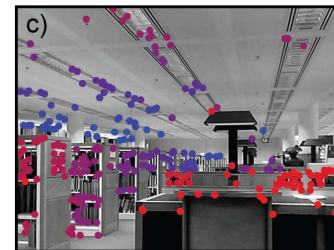


$$Z = \frac{fT}{x_r - x_l} = \frac{fT}{d}$$

5

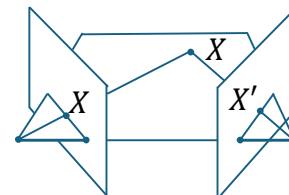
Story So Far

- The problem: compute sparse depth with **unaligned** cameras
- Again perform stereo calibration
 - Intrinsics and extrinsics known
- Depth computation is easy
 - If Pair-wise correspondence is known



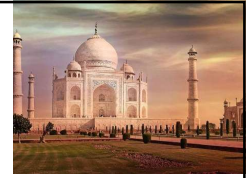
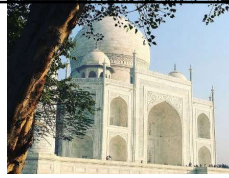
Sparse Depth

Solve for coefficients a_0, b_0, c_0
 $T + b_0 R^T p_r = a_0 p_l + c_0 (p_l \times R^T p_r)$

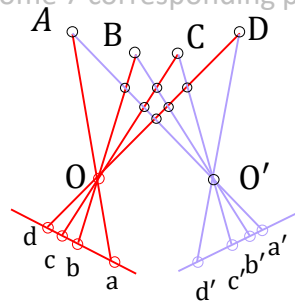


6

Key Challenge



- Given “casual” left and right images, how can we (efficiently) find the correspondence for random points in the scene?
 - Unaligned cameras
 - The “fundamental” matrix is an important step
 - (if we are confident on some 7 corresponding points)



Nothing ==
casual picture

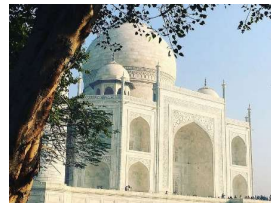
7

Overview

- Some fun stuff
 - Basics, buzzwords, and terminology on stereo
- The fundamental matrix F
 - Computing ‘K, R and T parameters’ from F
- The essential matrix E



Stereo parameters: K
and (R, T) known



Neither K nor (R, T) known

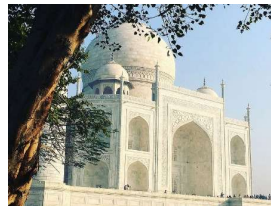
8

Overview

- Some fun stuff
 - Basics, buzzwords, and terminology on stereo
- The fundamental matrix F
 - ~~Computing 'K, R, T parameters' from F~~
 - ~~Rectification for solving correspondence~~
- The essential matrix E



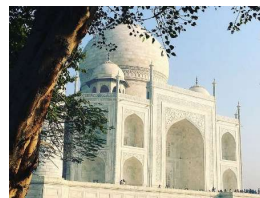
Stereo parameters: K
and (R, T) known



9

Story So Far

- The problem: compute sparse depth
 - We will **assume** intrinsics
 - We will **not assume** extrinsics
 - We will not assume correspondence
 - We will not assume aligned camera
- Concept of essential matrix



K known, (R, T) unknown

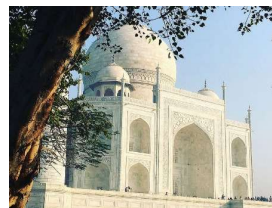
10

Overview

- Some fun stuff
 - Basics, buzzwords, and terminology on stereo
 - The fundamental matrix F
 - ~~Computing 'K, R and T parameters' from F~~
 - The essential matrix E
 - We know K (it's our camera!)
 - but we don't know R and T
- 



Stereo parameters: K and (R, T) known



11

Cameras With Known K

- If K is known, we can invert the image coordinate x to get $\hat{w} = K^{-1}x$
 - So-called normalized coordinate
- If we use the original non-normalized coordinates and $x'^T F x = 0$ we get relationship between F and K
 - $E = K'^T F K$ is called the essential matrix
 - $\hat{w}'^T E \hat{w} = 0$ (using normalized coordinates)

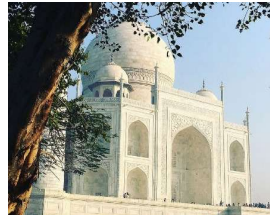
$$K \begin{bmatrix} \hat{w} \end{bmatrix} = K^{-1} \begin{bmatrix} x \end{bmatrix} \begin{bmatrix} \end{bmatrix}_{3 \times 1} \begin{bmatrix} \end{bmatrix}_{3 \times 1} \begin{bmatrix} \end{bmatrix}_{3 \times 1} p=0$$

$$\begin{aligned} X' &= K \hat{w} \\ X' E^T X &= 0 \\ (K \hat{w})^T (K w) &= 0 \\ \hat{w}^T (K^T E K) w &= 0 \\ \hat{w}^T E(w) &= 0 \end{aligned}$$

12

Cameras With Known K

- If K is known, and a few points are known, we know E
- With E , it is possible to get relative camera orientation, i.e., R, T
 - Once we know K, R, T we can triangulate
- How is it possible?
 - Properties of E

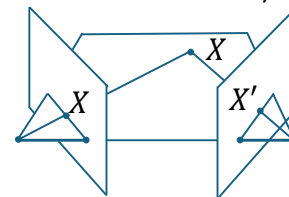


K known, (R, T) unknown

13

Essential Matrix Properties

- $E = K'^T F K$ and we can show that $E = t_{\times} R$
- E has five degrees of freedom since $E = SR$
 - 3 each for rotation and translation minus one for homogeneity
- S is the skew symmetric form of t
 - Rank of E is 2
 - Further, the two singular values are equal (and the third is zero as before)
- Now we know K, R, T
 - we can triangulate



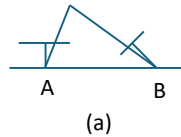
14

Depth with Essential Matrix

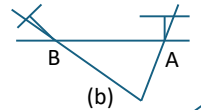
$$\begin{cases} ax^2 + bx + c = 0 \\ x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{cases}$$

- After reconstruction of a 3D point X , the four cases will result in only one case with the point in front of the cameras

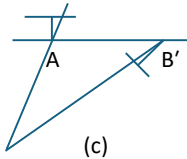
(a) Reconstructed point is in front of both cameras



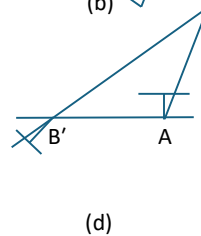
(b) Baseline reversal w.r.t. (a)



(c) Camera A finds that the reconstructed point is behind!



(d) Camera B finds that the reconstructed point is behind!



17

Summary: known intrinsic parameters

- The extrinsic parameters of the stereo system (i.e., rotation and translation between the optical axes of the two cameras) are unknown.
- We can estimate only the direction of the baseline vector (i.e., translation direction T) and not its magnitude.
- We can estimate the 3D coordinates of the points only up to an unknown scale.

The **essential matrix** E gives the relationship between the corresponding points measured in **camera coordinates**.

The **fundamental matrix** F gives the relationship between the corresponding points measured in **homogeneous** coordinates with the x and y components measured in the **pixel coordinate system**.

19

Summary

- How to do reconstruction if parameters are known
 - Unaligned but known cameras
- Some fun stuff!!
- Basics, buzzwords, and terminology on stereo
- The fundamental matrix F
 - Computing 'parameters' from F
- The essential matrix E
 - Computing 'parameters' from E